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Studierichting: Informatica

Oefeningenreeks 6E nr. 1.6

$$T_4\left(f, \frac{\pi}{2}\right)(x) \text{ voor } f(x) = \sin x$$

$$T_n(f, a)(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k$$

$$T_4\left(f, \frac{\pi}{2}\right)(x) = \sum_{k=0}^4 \frac{f^{(k)}\left(\frac{\pi}{2}\right)}{k!} \left(x - \frac{\pi}{2}\right)^k$$

$$f^{(1)}(x) = \cos x \Rightarrow f^{(1)}\left(\frac{\pi}{2}\right) = 0$$

$$f^{(2)}(x) = -\sin x \Rightarrow f^{(2)}\left(\frac{\pi}{2}\right) = -1$$

$$f^{(3)}(x) = -\cos x \Rightarrow f^{(3)}\left(\frac{\pi}{2}\right) = 0$$

$$f^{(4)}(x) = \sin x \Rightarrow f^{(4)}\left(\frac{\pi}{2}\right) = 1$$

$$\begin{aligned} T_6\left(f, \frac{\pi}{2}\right)(x) &= \frac{f^0\left(\frac{\pi}{2}\right)}{0!} \left(x - \frac{\pi}{2}\right)^0 + \frac{f^{(1)}\left(\frac{\pi}{2}\right)}{1!} \left(x - \frac{\pi}{2}\right)^1 + \frac{f^{(2)}\left(\frac{\pi}{2}\right)}{2!} \left(x - \frac{\pi}{2}\right)^2 + \\ &\frac{f^{(3)}\left(\frac{\pi}{2}\right)}{3!} \left(x - \frac{\pi}{2}\right)^3 + \frac{f^{(4)}\left(\frac{\pi}{2}\right)}{4!} \left(x - \frac{\pi}{2}\right)^4 \\ &= 1 - \frac{1}{2} \left(x - \frac{\pi}{2}\right)^2 + \frac{1}{24} \left(x - \frac{\pi}{2}\right)^4 \end{aligned}$$

References